SCALING DATABASE SYSTEMS

Prerequisites

Winter Term 2026/27 · Chair of Scalable Database Systems
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1 Required Skills

In this course, you will build miniHive, a minimalistic version of Hive. You will make **extensive use of the command line (terminal) and Python**. Further, there will be a Docker container for you to test your implementation.

For the first exercise session, please ensure that you have a working Docker setup (see Section 2) and a good understanding of the following:

- Basic usage of the Bash command line interface in Linux. This is important, the **miniHive container does not have a graphical user interface!**
- Concept and basic usage of Git (version control system).
- Basic usage of SSH (how to connect to a remote machine).

If you are not familiar with these topics, there are several free courses and documentation to catch up on the basics.

- Bash: `https://www.youtube.com/playlist?list=PLAwxTw4SYaPlBwWBVoIWKfSWxzuhfSW-D`
- Git: `https://www.udacity.com/course/version-control-with-git--ud123`
- SSH: `https://www.fim.uni-passau.de/en/it-services/remote-access/remote-login-via-ssh`

2 Docker Installation

To install Docker on your machine, follow the instructions on `https://docs.docker.com/get-docker`. Make sure to run the hello-world image as described in the Docker documentation to verify that your Docker environment is installed correctly.

2.1 Hardware Resources

We recommend at least 8 GiB of main memory for running the miniHive Docker container. You may use our *Docker server* if you cannot run the miniHive Docker container on your own device. Please see the sheet “Docker Server Instructions” for more information.

3 Questions?

Use the course forum in Stud.IP for questions and technical problems. Before posting a new question, check whether your question or problem has already been answered or solved.

When asking for help with technical problems, please provide enough relevant details, such as your operating system, the failed command, etc.

Note: As every machine has a different environment (operating system, applications installed, etc.) and different hardware, we can only provide limited support for technical problems on your own device.